

algorithms produce an essentially unbiased DOA estimate regardless of polarization.

Fig. 6 presents a "plan view" of the "ridge lines" for the surfaces of Figs. 3–5 with a superimposed ridge line for the DPMUSIC spectrum. The curves indicate the DOA estimates versus polarization for the three cases. Clearly BS produces biased DOA estimates except for polarizations corresponding to the calibration data. By contrast the ML algorithms and DPMUSIC produce essentially unbiased DOA estimates for all slant polarizations.

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The Variance of Multitaper Spectrum Estimates for Real Gaussian Processes

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Abstract—Multitaper spectral estimation has proven very powerful as a spectral analysis method wherever the spectrum of interest is detailed and/or varies rapidly with a large dynamic range. In his original paper D. J. Thomson gave a simple approximation for the variance of a multitaper spectral estimate which is generally adequate when the spectrum is slowly varying over the taper bandwidth. We show that near zero or Nyquist frequency this approximation is poor even for white noise and derive the exact expression of the variance in the general case of a stationary real-valued time series. This expression is illustrated on an autoregressive time series and a convenient computational approach outlined. It is shown that this multitaper variance expression for real-valued processes is not derivable as a special case of the multitaper variance for complex-valued, circularly symmetric processes, as previously suggested in the literature.

I. INTRODUCTION

The technique of multitaper spectral analysis formulated by Thomson [1] is a popular method to use whenever the spectrum of interest is detailed and/or varies rapidly with a large dynamic range. The tapers are designed for suppression of side-lobe leakage, while the multitaper aspect ensures stable estimates. There is a growing number of substantive scientific applications of the technique [2]–[9]. Bronez [10] extended the technique to irregularly sampled processes, and—based upon performance bounds—argued in [11] that the technique has advantages over Weighted Overlapped Segment Averaging (WOSA).

In this paper we look at the variance of multitaper spectrum estimates for real, (and also complex), Gaussian processes. In Section II we derive an exact variance expression, and illustrate it in Section III by application to real-valued white noise and autoregressive processes.

II. DERIVATION OF THE VARIANCE

A. Real Process Case

Let $\{X_t\}$ be a discrete, real-valued, purely nondeterministic stationary process with mean zero, autocovariance sequence $\{R_k\}$, and spectral density function $S(f)$. Given $X_1, \dots, X_N$ the basic multitaper estimate of $S(f)$ is given in [1] as

$$\hat{S}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N h_{t,k}(N, W) X_t e^{-i2\pi f t} \right|^2 \quad (1)$$

where $\{h_{t,k}(N, W)\}$, $k = 0, \dots, K-1$ are a set of orthonormal tapers which depend on the sequence length N and the half-bandwidth W . The elements of the taper $\{h_{t,k}(N, W)\}$, $t = 1, \dots, N$ are those of a discrete prolate spheroidal sequence (dpss), of order k . The tapers are eigenvectors of a matrix eigenproblem, and each has an

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associated eigenvalue $\lambda_k(N, W)$. The side-lobe leakage of each taper is proportional to $1 - \lambda_k(N, W)$, and hence only those tapers for which $\lambda_k(N, W)$ is close to unity are generally useful. The number of tapers used, K is hence typically $2NW - 1$ or $2NW - 2$ (see [1]). For notational convenience we henceforth drop the explicit dependence of the tapers on (N, W) .

Let us now write

$$\hat{S}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \hat{S}_k(f) \quad (2)$$

where

$$\hat{S}_k(f) = \left| \sum_{t=1}^N h_{t,k} X_t e^{-i2\pi f t} \right|^2 \quad (3)$$

is called the k th eigenspectrum. The variance of this estimator is given by

$$\text{var}\{\hat{S}(f)\} = \frac{1}{K^2} \sum_{j=0}^{K-1} \sum_{k=0}^{K-1} \text{cov}\{\hat{S}_j(f), \hat{S}_k(f)\}. \quad (4)$$

The assumption of uncorrelated eigenspectra, which follows from the orthogonality of the tapers [1] if the spectrum is not rapidly varying over $[f - W, f + W]$, gives $\text{var}\{\hat{S}(f)\} = (1/K^2) \sum_{k=0}^{K-1} \text{var}\{\hat{S}_k(f)\}$. For a Gaussian time series we thus see that each additional (chi-squared) term in (2) contributes two degrees of freedom to the spectrum estimate, giving a total of $2K$ degrees of freedom at any frequency (except close to 0 and Nyquist). Combining this with the asymptotic expression for direct spectral estimators, $\text{var}\{\hat{S}_k(f)\} \approx S^2(f)$, it follows that $\hat{S}(f)$ is approximately distributed as $(1/2K)S(f)\chi_{2K}^2$, or equivalently

$$\text{var}\{\hat{S}(f)/S(f)\} = 1/K. \quad (5)$$

This approximate result from [1] assumes a spectrum not rapidly varying over $[f - W, f + W]$. However, the multitaper method will often be applied in the cases of high dynamic range, rapidly varying spectra, so this assumption is not ideal. We can calculate a generally valid exact result by evaluating the covariances in (4). To do so we need an expression for $\text{cov}\{Z_1 Z_2, Z_3 Z_4\}$ where (Z_1, Z_2, Z_3, Z_4) have the multivariate complex Gaussian distribution with zero mean and arbitrary covariance matrix (in the sense of not being necessarily circularly symmetric). As there are differing conventions for complex variables we firstly define $\text{cov}\{Z_j, Z_k\} = E\{Z_j^* Z_k\} - E\{Z_j^*\} E\{Z_k\}$ so that

$$\begin{aligned} \text{cov}\{Z_1 Z_2, Z_3 Z_4\} \\ = E\{Z_1^* Z_2^* Z_3 Z_4\} - E\{Z_1^* Z_2^*\} E\{Z_3 Z_4\}. \end{aligned} \quad (6)$$

The complex version of the Isserlis theorem gives [12, p. 27]

$$\begin{aligned} E\{Z_1^* Z_2^* Z_3 Z_4\} &= E\{Z_1^* Z_2^*\} E\{Z_3 Z_4\} \\ &\quad + E\{Z_1^* Z_3\} E\{Z_2^* Z_4\} \\ &\quad + E\{Z_1^* Z_4\} E\{Z_2^* Z_3\}. \end{aligned}$$

Thus from (6)

$$\begin{aligned} \text{cov}\{Z_1 Z_2, Z_3 Z_4\} &= \text{cov}\{Z_1, Z_3\} \text{cov}\{Z_2, Z_4\} \\ &\quad + \text{cov}\{Z_1, Z_4\} \text{cov}\{Z_2, Z_3\}. \end{aligned} \quad (7)$$

Now let $J_j(f) = \sum_{u=1}^N h_{u,j} X_u \exp(-i2\pi f u)$ and $J_k(f) = \sum_{t=1}^N h_{t,k} X_t \exp(-i2\pi f t)$. Note that $E\{J_j(f)\} = E\{J_k(f)\} = 0$.

Then from (3) and (7)

$$\begin{aligned} \text{cov}\{\hat{S}_j(f), \hat{S}_k(f)\} &= \text{cov}\{|J_j(f)|^2, |J_k(f)|^2\} \\ &= \text{cov}\{J_j(f) J_j^*(f), J_k(f) J_k^*(f)\} \\ &= \text{cov}\{J_j(f), J_k(f)\} \\ &\quad \cdot \text{cov}\{J_j^*(f), J_k^*(f)\} \\ &\quad + \text{cov}\{J_j(f), J_k^*(f)\} \\ &\quad \cdot \text{cov}\{J_j^*(f), J_k(f)\} \\ &= E\{J_j^*(f) J_k(f)\} E\{J_j(f) J_k^*(f)\} \\ &\quad + E\{J_j^*(f) J_k^*(f)\} E\{J_j(f) J_k(f)\} \\ &= |E\{J_j^*(f) J_k(f)\}|^2 + |E\{J_j(f) J_k(f)\}|^2. \end{aligned}$$

For the first of these terms

$$|E\{J_j^*(f) J_k(f)\}|^2 = \left| \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi f(t-u)} \right|^2.$$

For the second term

$$|E\{J_j(f) J_k(f)\}|^2 = \left| \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi f(t+u)} \right|^2.$$

Hence, finally from (4)

$$\begin{aligned} \text{var}\{\hat{S}(f)\} &= \frac{1}{K^2} \left\{ \sum_{j=0}^{K-1} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi f(t-u)} \right|^2 \right. \\ &\quad \left. + \sum_{j=0}^{K-1} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi f(t+u)} \right|^2 \right\}. \end{aligned} \quad (8)$$

This expression is very valuable in three respects:

1. It is readily computable at the Fourier frequencies for a model with known autocovariance sequence using a series of fast Fourier transforms (FFT's). First, use a forward FFT to give

$$Y_{u,k}(f_\ell) = \frac{1}{N} \sum_{t=1}^N h_{t,k} R_{t-u} e^{-i2\pi f_\ell t}$$

where $f_\ell = \ell/N$. Then apply an inverse FFT to $\{h_{u,j} Y_{u,k}(f_\ell)\}$ and multiply by N to obtain

$$\begin{aligned} N \left\{ \sum_{u=1}^N h_{u,j} Y_{u,k}(f_\ell) e^{i2\pi u f_\ell} \right\} \\ = \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi(t-u)f_\ell}. \end{aligned}$$

For the second term, apply a forward FFT to $\{h_{u,j} Y_{u,k}(f_\ell)\}$ and multiply by N^2 to

$$\begin{aligned} N^2 \left\{ \frac{1}{N} \sum_{u=1}^N h_{u,j} Y_{u,k}(f_\ell) e^{-i2\pi u f_\ell} \right\} \\ = \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} R_{t-u} e^{-i2\pi(t+u)f_\ell}. \end{aligned}$$

Note that (7.1) of [1] is the frequency domain equivalent of (8), but whereas the former provides more insight, (8) provides the computational route.

2. It can be used to assess the importance of the (off-diagonal) covariances by computing the variance with and without these terms.
3. We can see how the variance at different frequencies is related to the number of tapers.

B. Complex Process Case

Consider now a Gaussian and circularly-symmetric complex stationary process, $\{Z_t\}$ in (1). Using the fact that circular symmetry ensures that the real and imaginary parts of the process have the same variance and are uncorrelated, Bronez [10] derived the variance of this spectral estimator, which can be rewritten as

$$\text{var}\{\hat{S}(f)\} = \frac{1}{K^2} \left\{ \sum_{j=0}^{K-1} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N \sum_{u=1}^N h_{u,j} h_{t,k} V_{t-u} e^{-i2\pi(t-u)f} \right|^2 \right\} \quad (9)$$

which is identical to the first term in (8) except that the real autocovariance sequence $\{R_k\}$ in (8) has been replaced by its complex equivalent $\{V_k\}$. For real processes, Bronez [10, Appendix A] claimed that "the variance expression for real-valued Gaussian processes differs by a factor of two from the expression for complex-valued, circularly symmetric Gaussian processes." Comparing (8) and (9) we see that this cannot be correct because the second term in (8) involving $\exp[-i2\pi(t+u)f]$ is not present in (9).

III. EXAMPLE VARIANCES FOR MULTITAPER ESTIMATION

A. White Noise

The first example of the use of the variance expression in (8) is for a real-valued white noise process with variance σ^2 of unity. This is the ideal case where the spectrum does not change rapidly—in fact it changes not at all—over $[f - W, f + W]$; side-lobe leakage is not an issue in this case. For $\{X_t\}$ white noise, $S(f) = \sigma^2$ for all frequencies and $R_{t-u} = \sigma^2$ when $t = u$ and zero otherwise. Hence from (8)

$$\begin{aligned} \text{var}\{\hat{S}(f)/S(f)\} &= \frac{1}{K^2} \left\{ \sum_{k=0}^{K-1} \left[\sum_{t=1}^N h_{t,k}^2 \right]^2 \right. \\ &\quad \left. + \sum_{j=0}^{K-1} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N h_{t,j} h_{t,k} e^{-i4\pi t f} \right|^2 \right\} \\ &= \frac{1}{K} + \frac{1}{K^2} \sum_{k=0}^{K-1} \left| \sum_{t=1}^N h_{t,k} e^{-i4\pi t f} \right|^2 \\ &\quad + \frac{1}{K^2} \sum_{\substack{j,k=0 \\ j \neq k}}^{K-1} \left| \sum_{t=1}^N h_{t,j} h_{t,k} e^{-i4\pi t f} \right|^2 \quad (10) \end{aligned}$$

the first term following from the orthonormality of the tapers. The first term in (10) gives agreement with (5). The last term in (10) is the effect of the covariances. Taking $N = 50$ and $W = 0.05$, the variance in (10) was calculated, including and excluding the covariances, for the number of tapers K ranging from 1 to 6, and is given in Fig. 1(a) and (b) for $f = 0.02$ and $f = 0.24$, respectively. The variance has first been divided by the true spectrum value squared (here just $\sigma^4 = 1$), and then plotted on a log scale. (Note that because of the symmetries in this case, the variance at f is the same as at $0.5 - f$.) Let us look first at Fig. 1(a). The nominal $1/K$ approximation is plotted. Also plotted is the variance from (8) or (10) excluding covariances, i.e., the sum of the first two terms in (10), and the variance from (8) or (10) including covariances, i.e., all terms in (10). We see that the second term in (10) increases the $1/K$ line by a small amount, while the third term (the covariances) leads to a much more substantial increase in the variance estimate from the nominal $1/K$. Turning now to Fig. 1(b), the three cases cannot be distinguished. (In fact for six tapers the first sign of a difference is apparent, with a slightly larger variance value when covariances are included—this difference would grow slowly with

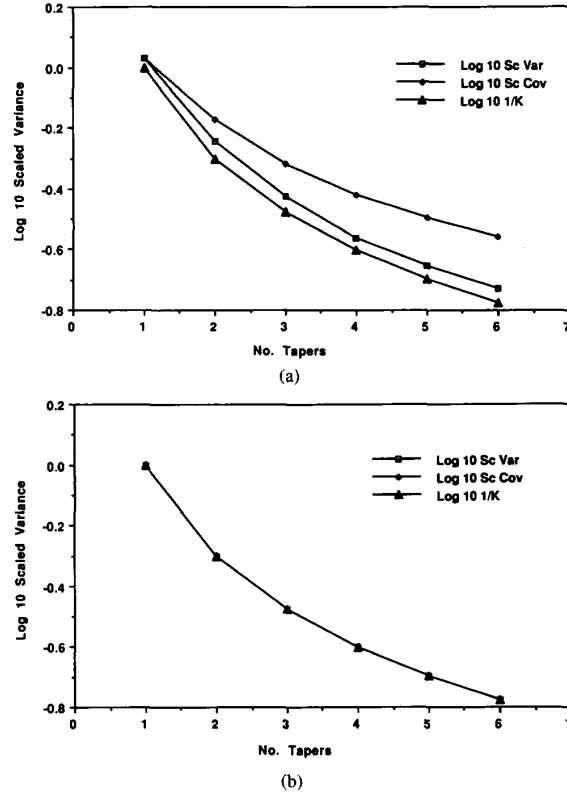


Fig. 1. Exact scaled variance of multitaper spectral estimator, including (Cov) and excluding (Var) covariances, for white noise at (a) $f = 0.02$, and (b) $f = 0.24$. Also shown is the nominal $1/K$ curve.

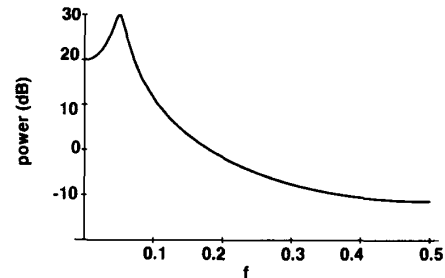


Fig. 2. Power spectrum of AR(2) process with unity innovations variance and autoregressive parameters $\phi_1 = 1.8$ and $\phi_2 = -0.9$.

increasing numbers of tapers.) Generally for white noise, as illustrated in these two figures, the frequency dependent terms in (10) are much more important for f close to 0 or 0.5, and much less so in the middle range of frequencies.

B. Autoregressive Process

To further illustrate the use of the variance expression in (8) we take a real-valued second-order autoregressive time series model with high dynamic range and rapidly varying spectra, given by $X_t = 1.8X_{t-1} - 0.9X_{t-2} + \epsilon_t$ where $\{\epsilon_t\}$ is a zero-mean white noise process with unit variance. The spectrum is shown in Fig. 2; it has a peak at about $f = 0.05$. Suppose we take $N = 50$, and $W = 0.04$, i.e., $2NW = 4$. These choices of N and W mean that

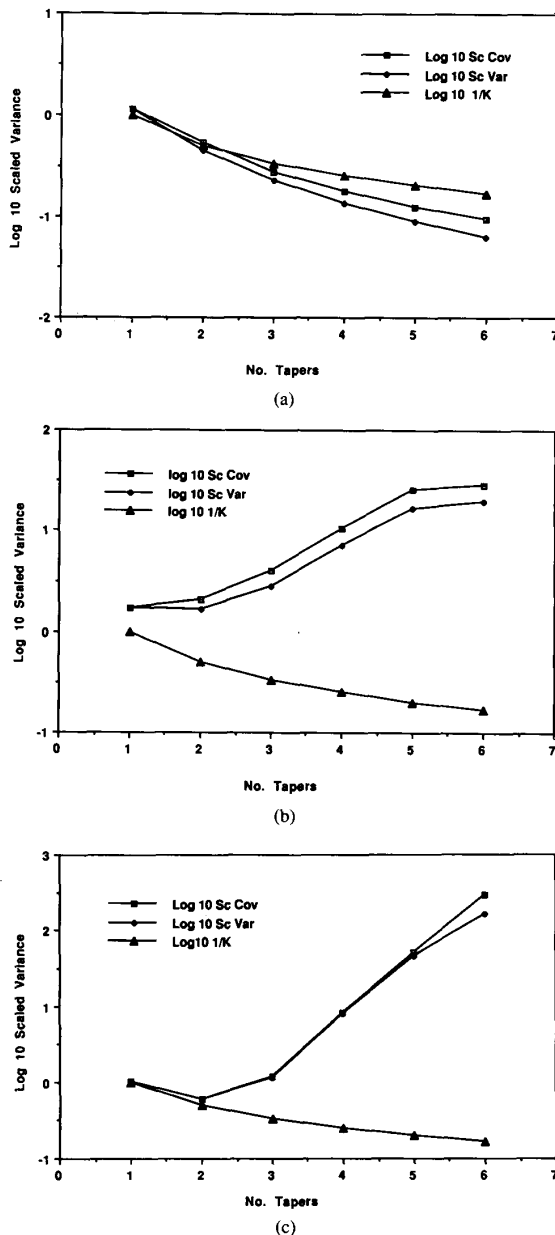


Fig. 3. Exact scaled variance of multitaper spectral estimator, including (Cov) and excluding (Var) covariances, for AR(2) process at (a) $f = 0.04$, (b) $f = 0.1$, and (c) $f = 0.24$. Also shown is the nominal $1/K$ curve.

we should only expect good multitaper behavior when using no more than two or three tapers since the number of tapers should be chosen to be $K \leq 2NW - 1$, i.e., two or three. The approximate result in (5) assumes the true spectrum $S(f)$ does not vary rapidly over $[f - W, f + W] = [f - 0.04, f + 0.04]$. From Fig. 2 we can see that the spectrum does vary rapidly over such an interval, especially close to the peak at $f = 0.05$.

In Fig. 3(a) we give the variance in (8) including and excluding the covariances, for the number of tapers K ranging from 1 to 6, at frequency $f = 0.04$. The range of the spectrum in the frequency interval $[0.0, 0.8]$ is 13 dB. The variance has again been divided by

the true spectrum value squared, and then plotted on a log scale. Also shown is the $1/K$ curve of (5). The frequency $f = 0.04$ is very close to the spectral peak, and all three curves are in close agreement for the first two or three tapers, agreeing with the simple theory. This good behavior is presumably due to the fact that near the peak of the spectrum side-lobe leakage is not a problem (power can only leak out), and locally the spectrum is close to symmetric. In Fig. 3(b) we look at the frequency $f = 0.1$; the range of the spectrum in the interval $[0.06, 0.14]$ is high at 21 dB. We can see that the exact variance from (8) is quite unlike the approximation in (5). This is because the spectrum is varying rapidly around $f = 0.1$ and the spectrum estimate subject to side-lobe leakage. Note that the exact variance (including the covariances) increases as soon as more than just one taper is used. Away from the peak, at $f = 0.24$, expressions (5) and (8) give very close agreement (Fig. 3(c)) for the first two tapers and the exact variance decreases: the spectrum is not varying too rapidly here, and the spectral range in the interval $[0.2, 0.28]$ is low at 5 dB. However, in line with theory, additional tapers cause a rapid rise in variance, especially with four or more tapers, because the lower power area ($f > 0.2$) is very prone to leakage from the spectral peak.

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